AI ASSITIVE SMART CANE

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Department of Electronics and Communication Engineering(ECE)

CERTIFICATE

This is to certify that this is the project report entitled **"AI ASSISTIVE SMART CANE "**submitted by **Y.PRANEETH RAM(2411EC010070), S.VIVEK(2411EC010112), N.VISHNU PRABATH** B.Tech III year – I Semester, Department of Electronics and Communication Engineering(ECE), during the year 2025-26. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project report **"AI ASSISTIVE SMART CANE",** has been carried out by us and this work has been submitted to the **Department of Electronics and Communication Engineering (ECE), Malla Reddy University, Hyderabad**. We further declare that this project work has not been submitted in full or part for the award of any other degree in any other educational institutions.

Place:

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ABSTRACT

Our project is called **AI Assistive Smart Cane**, and we made it to help blind people while walking. The main idea is very simple: if there is something in front of the person, the cane should warn them before they hit it. We are using an **ESP32** to control the whole system. The ultrasonic sensor checks the distance in front of the cane. When an object comes within **30 cm**, the buzzer gives two beeps, and if it comes closer than **27 cm**, the buzzer keeps beeping continuously. We also have a **water sensor** at the bottom of the cane, which can detect water or a wet surface. When water is detected, the vibration motors give a vibration that the person can feel through the handle. We are also using an **MPU6050** to check the movement and position of the cane. We chose these components because they are simple, affordable, and easy to control using the ESP32. Our main goal is to make a normal cane more useful by giving the user **sound and vibration warnings** about things they may not be able to see.

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CHAPTER - 1

INTRODUCTION

INTRODUCTION

The AI Assistive Smart Cane is designed to help blind and visually impaired people walk more safely and independently. Traditional walking canes have limitations in detecting obstacles before reaching them and identifying wet surfaces.

To overcome these limitations, our project uses an ESP32 microcontroller to control different sensors and alert systems. An ultrasonic sensor detects obstacles in front of the user and activates a buzzer when an object comes within 30 cm. If the object comes closer than 27 cm, the buzzer gives continuous beeping. A water sensor detects wet surfaces and activates vibration motors to alert the user through touch. An MPU6050 sensor is also used to monitor the movement and orientation of the cane.

The main aim of this project is to improve a traditional walking cane using affordable electronic components and provide better safety through sound and vibration alerts.

1.1 Background of the Problem

Blind and visually impaired people face many difficulties while walking independently because they cannot easily identify obstacles and wet surfaces in their surroundings. Traditional walking canes help detect obstacles through physical contact, but they cannot always provide early warnings or detect water on the ground. To overcome these limitations, the AI Assistive Smart Cane is developed using an ESP32 microcontroller, ultrasonic sensor, water sensor, buzzer, vibration motors, and MPU6050. The system detects nearby obstacles and wet surfaces and alerts the user through sound and vibrations. It also provides different warning signals based on the distance of the obstacle. The main aim of this project is to improve the safety of visually impaired people and make their daily movement easier and more independent.

1.2 Motivation and Need for the Project

The main motivation behind this project is to help blind and visually impaired people move safely and independently in their daily lives. Traditional walking canes have limitations in detecting obstacles at a distance and identifying wet surfaces. To overcome these problems, we developed the AI Assistive Smart Cane using an ESP32 and different sensors. It provides sound alerts for nearby obstacles and vibration alerts when water is detected. This project aims to reduce the risk of accidents, improve user safety, and provide an affordable smart solution for visually impaired people. It also helps users become more confident while walking in different surroundings. By using

simple and low-cost electronic components, the system provides an improved alternative to traditional walking canes.

1.3 Problem Statement

Blind and visually impaired people face difficulties while walking because they cannot easily identify obstacles and wet surfaces in their path. Traditional walking canes provide only limited assistance and may not give early warnings about nearby objects or water on the ground. This can increase the risk of accidents and injuries. Therefore, there is a need for an affordable smart walking cane that can detect obstacles and water and alert users through sound and vibrations. Our AI Assistive Smart Cane uses an ESP32, ultrasonic sensor, water sensor, buzzer, and vibration motors to provide warnings and improve the safety and independence of visually impaired people.

1.4 Objectives of the Project

The main objective of the AI Assistive Smart Cane is to help blind and visually impaired people walk safely and independently. The project aims to detect obstacles in front of the user using an ultrasonic sensor and provide buzzer alerts based on the distance. It also aims to detect water or wet surfaces using a water sensor and alert the user through vibration motors. The MPU6050 sensor is used to monitor the movement and orientation of the cane. The ESP32 controls all the connected components. Overall, the project focuses on developing an affordable, user-friendly, and reliable smart walking cane to improve the safety and confidence of visually impaired people.

1.5 Scope of the Project

The scope of the AI Assistive Smart Cane is to improve the safety and mobility of blind and visually impaired people during walking. The system detects nearby obstacles using an ultrasonic sensor and provides sound alerts through a buzzer. It also detects water or wet surfaces using a water sensor and alerts the user through vibration motors. The ESP32 controls all the sensors and warning devices, while the MPU6050 monitors the movement and orientation of the cane. This project focuses on providing an affordable and easy-to-use solution that can reduce walking difficulties and help visually impaired people move with greater confidence and independence.

1.6 Organization of the Report

This report presents the design and development of the AI Assistive Smart Cane project in a systematic manner. Chapter 1 introduces the project, including its background, motivation, problem statement, objectives, and scope. Chapter 2 discusses the literature survey and existing mobility assistance systems. Chapter 3 explains the proposed system, its architecture, and the hardware and software requirements. Chapter 4 describes the system implementation, sensor integration, and machine-learning techniques. Chapter 5 presents the testing procedures, results, and observations. Finally, Chapter 6 provides the conclusion and future scope of the project.

CHAPTER – 2

LITERATURE AND EXISTING SYSTEM

2.1 Overview of Previous Works

Several smart cane systems have been developed to assist visually impaired individuals in detecting obstacles and navigating their surroundings. Earlier systems mainly used ultrasonic sensors to measure the distance between the user and nearby objects, providing alerts through buzzers or vibration motors. Some advanced systems introduced GPS and IoT technologies for location tracking and emergency notifications. However, many existing systems rely on fixed distance thresholds and have limited ability to distinguish between different types of environmental hazards. Recent developments have explored Artificial Intelligence and machine-learning techniques to improve obstacle detection and classification. The proposed AI Assistive Smart Cane builds on these approaches by combining multiple sensors, lightweight machine learning, vibration-based feedback, and IoT connectivity to develop an affordable and intelligent mobility assistance prototype..

2.2 Limitations in Existing Systems

Existing smart cane systems have several limitations in detecting and identifying environmental hazards. Many traditional systems rely on ultrasonic sensors and fixed distance thresholds, which may detect nearby obstacles but cannot accurately distinguish between different types of hazards. Some systems provide only basic buzzer alerts, making it difficult for users to understand the location or type of obstacle. Advanced smart canes with GPS and IoT features can be expensive and require additional hardware or internet connectivity. Certain systems also have limited capabilities in detecting low obstacles, sudden drops, steps, and water hazards. These limitations highlight the need for a low-cost intelligent smart cane that combines multiple sensors, machine-learning techniques, and vibration-based feedback to improve environmental awareness and mobility assistance.

2.3 Research Gap Identified

Existing smart cane systems mainly focus on basic obstacle detection using ultrasonic sensors and fixed distance thresholds. Many systems have limited capabilities in identifying different environmental hazards, such as low obstacles, steps, sudden drops, and water hazards. Some advanced systems use Artificial Intelligence and IoT technologies but may involve higher costs and complex hardware. There is a need for an affordable and intelligent mobility assistance system that can classify multiple hazards and provide clear, immediate feedback. The proposed AI Assistive Smart Cane aims to address this gap by integrating multiple sensors, lightweight machine-learning techniques, distinct vibration patterns, and optional IoT connectivity to support safer and more independent mobility for visually impaired individuals.

2.4 Summary

The literature survey highlights the development of various smart cane systems designed to assist visually impaired individuals in detecting obstacles and navigating safely. Existing systems mainly use ultrasonic sensors, buzzers, vibration motors, and IoT technologies for obstacle detection and user alerts. However, many systems have limitations in identifying different types of environmental hazards and providing accurate feedback. The identified research gap emphasizes the need for a low-cost intelligent system capable of detecting and classifying multiple hazards. The proposed AI Assistive Smart Cane addresses these challenges by integrating an ESP32 microcontroller, multiple sensors, lightweight machine-learning techniques, and vibration-based feedback. Additionally, IoT connectivity supports optional monitoring and emergency notifications. This approach aims to improve environmental awareness and provide an affordable mobility assistance solution for visually impaired individuals.

CHAPTER – 3

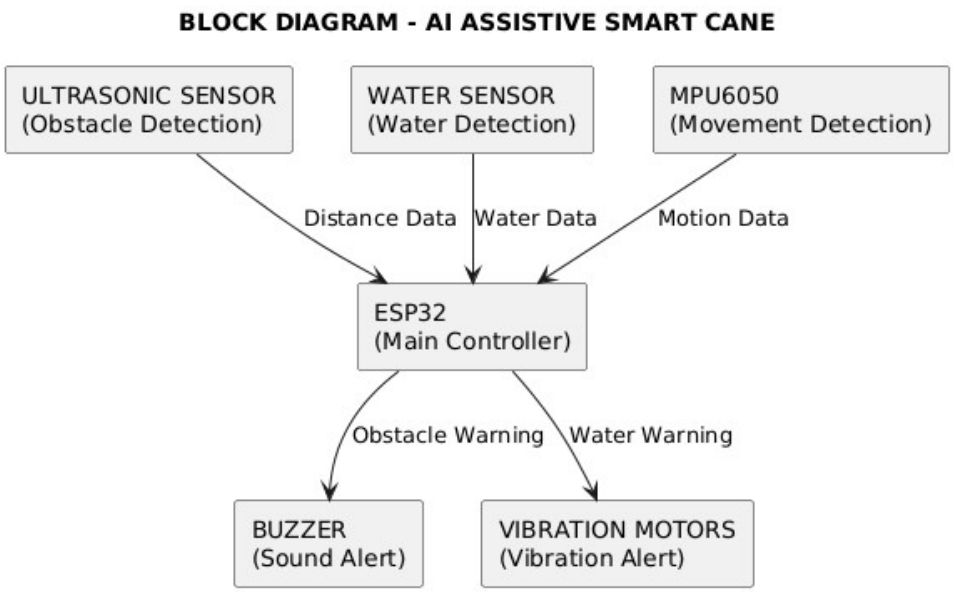
PROPOSED SYSTEM / METHODOLOGY

3.1 System Architecture / Block Diagram

The proposed AI Assistive Smart Cane is designed using an ESP32 microcontroller as the central processing unit, integrating multiple sensors, lightweight machine-learning techniques, vibration-based feedback, and IoT connectivity. The system architecture consists of four main sections: input sensors, data processing, hazard classification, and output alerts. The ultrasonic sensors continuously monitor the surrounding environment to detect obstacles and measure their distances. The MPU6050 accelerometer and gyroscope collect information about the movement, tilt, and orientation of the cane, which can assist in identifying possible steps and sudden drops. The water detection sensor identifies the presence of water on surfaces. All sensor readings are transmitted to the ESP32 microcontroller for processing and analysis.

The ESP32 processes the collected sensor data using a lightweight machine-learning model to classify different environmental conditions, including left-side obstacles, right-side obstacles, low obstacles, possible steps or sudden drops, and water hazards. Based on the identified hazard, the microcontroller activates vibration motors to generate distinct vibration patterns, allowing the user to recognize different conditions without visual assistance. The system is designed to minimize processing delays and provide quick feedback during movement. Additionally, the ESP32 supports Wi-Fi connectivity, enabling optional IoT-based device-status monitoring and emergency notifications through a connected device or platform.

The overall system follows a simple operational flow in which sensors continuously collect environmental information, the ESP32 analyzes the data using the machine-learning model, and the corresponding vibration alert is generated based on the detected hazard. IoT communication operates as an additional feature for remote monitoring and emergency support. The proposed architecture focuses on affordability, ease of implementation, and efficient operation using readily available hardware components. This integrated approach demonstrates how embedded systems, Artificial Intelligence, sensor technologies, and IoT can be combined to develop an intelligent mobility assistance prototype for visually impaired individuals.



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Fig. 3.1 Block Diagram

3.2 Working Principle

The AI Assistive Smart Cane works by continuously monitoring the surrounding environment using multiple sensors connected to an ESP32 microcontroller. The ultrasonic sensors detect nearby obstacles and measure their distances, while the MPU6050 accelerometer and gyroscope monitor the movement and orientation of the cane. The water detection sensor identifies the presence of water on surfaces. All sensor readings are collected and processed by the ESP32 microcontroller using lightweight machine-learning techniques to classify different environmental hazards.

Based on the sensor data, the system identifies conditions such as left-side obstacles, right-side obstacles, low obstacles, possible steps, sudden drops, and water hazards. Once a hazard is detected, the ESP32 immediately activates the vibration motors to generate a specific vibration pattern corresponding to the identified condition. These vibration alerts help users recognize potential dangers without requiring visual attention. Additionally, the ESP32 supports IoT connectivity through Wi-Fi for optional device-status monitoring and emergency notifications. The entire process operates continuously to provide quick feedback and support safer, more independent mobility for visually impaired individuals.

3.3 Detailed Explanation of Each Block/Component

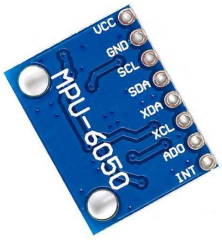
The AI Assistive Smart Cane consists of different hardware and software components that work together to detect obstacles and environmental hazards. The ESP32 microcontroller acts as the main processing unit, while various sensors collect information about the surroundings Each component plays an important role in providing intelligent mobility assistance to visually impaired.



1. **ESP32 Microcontroller:** The ESP32 is the central control unit of the system. It receives data from multiple sensors, processes the information using lightweight machine-learning techniques, and controls the vibration motors. It also supports Wi-Fi connectivity for IoT-based monitoring and emergency notifications.



2. **Ultrasonic Sensors:** Ultrasonic sensors detect nearby obstacles by transmitting ultrasonic waves and measuring their reflections. Multiple sensors are used to identify obstacles in different directions, including left-side, right-side, and low obstacles. The measured distances are sent to the ESP32 for analysis.



3. **MPU6050 Sensor:** The MPU6050 consists of an accelerometer and gyroscope that measure the movement, tilt, and orientation of the smart cane. This information helps the system recognize movement patterns and possible steps or sudden drops when combined with other sensor

readings.



4. **Water Detection Sensor:** The water detection sensor identifies the presence of water when its sensing surface comes into contact with it. The sensor sends information to the ESP32, which activates a specific vibration alert to warn the user.
5. **Machine-Learning Model:** The machine-learning model analyzes the collected sensor data and classifies different environmental hazards. It helps identify obstacles, possible steps, sudden drops, and water hazards without depending only on fixed distance thresholds.



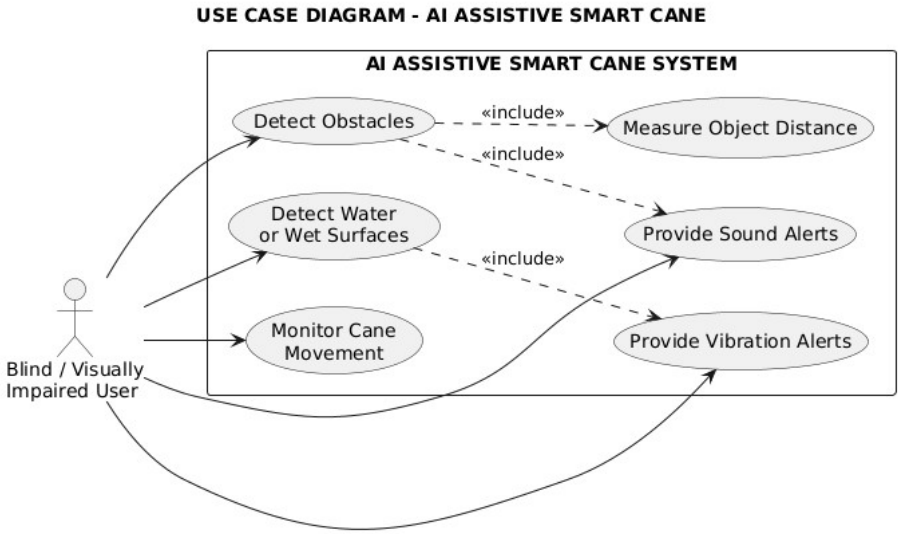
6. **Vibration Motors:** Vibration motors provide immediate haptic feedback to the user. Different vibration patterns are generated for different detected hazards, helping visually impaired individuals recognize potential dangers without visual assistance.

7. IoT Connectivity: The ESP32 provides Wi-Fi connectivity for optional remote monitoring and emergency notifications. This feature allows device information to be communicated through a connected device or IoT platform.

8. Power Supply: A battery and suitable voltage regulation module provide the required power to the ESP32, sensors, and vibration motors. The portable power supply allows the smart cane to operate without a continuous external power connection.

All these components work together to develop a low-cost intelligent mobility assistance prototype that aims to improve environmental awareness and support safer movement for visually impaired individuals.

3.3 UsecaseDiagram



3.4 Flowchart or Algorithm Steps



3.5 Hardware and Software Requirements

Hardware:

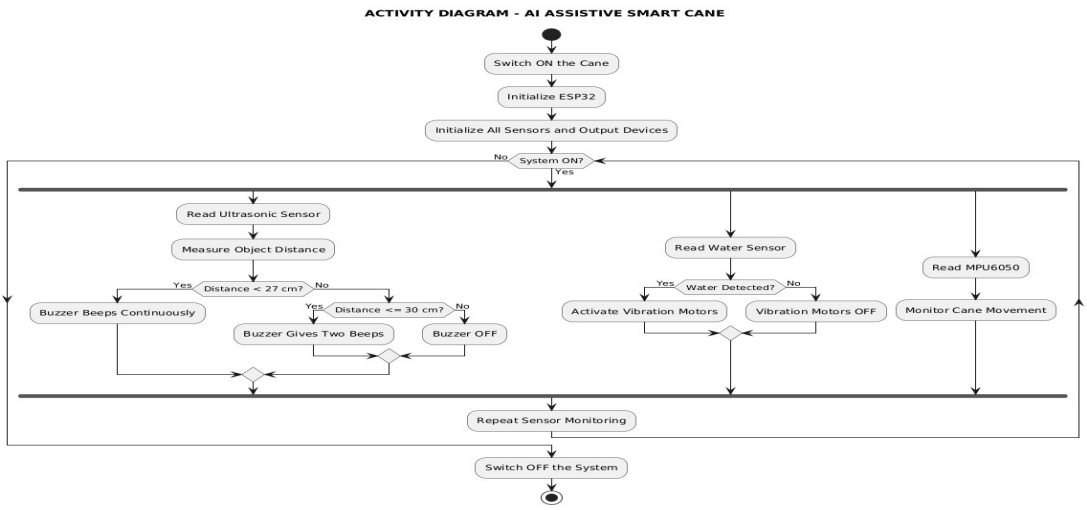
1	**ESP32 Microcontroller**	Main controller for sensor data processing, AI classification, and IoT connectivity.
2	**Ultrasonic Sensors (HC-SR04)**	Detect obstacles and measure distances in different directions.
3	**MPU6050 Sensor**	Measures cane movement, tilt, and orientation.
4	**Water Detection Sensor**	Detects water hazards on surfaces.
5	**Vibration Motors**	Provide different vibration patterns to alert the user.
6	**Rechargeable Battery**	Supplies power to the smart cane.
7	**Connecting Wires**	Connect sensors and other components to the ESP32.
8	**PCB / Breadboard**	Used for mounting and connecting electronic components.
9	**Smart Cane Structure**	Supports and holds all hardware components.
10	**Power Switch**	Turns the system ON and OFF.
11	**Motor Driver / Transistor Circuit**	Controls the vibration motors using ESP32 signals.

Main Hardware: ESP32, Ultrasonic Sensors, MPU6050, Water Detection Sensor, and Vibration Motors.

1. Relay module
2. wireless charging module
3. DC-DC booster converter

Software:

1	**Arduino IDE**	Used to write, compile, and upload code to the ESP32 microcontroller.
2	**Embedded C/C++**	Programming language used to control sensors and vibration motors.

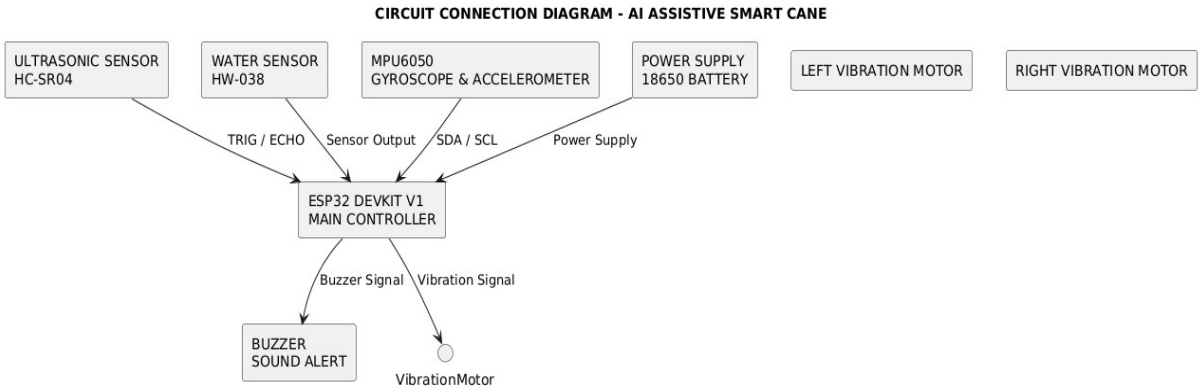


CHAPTER – 4

DESIGN AND IMPLEMENTATION

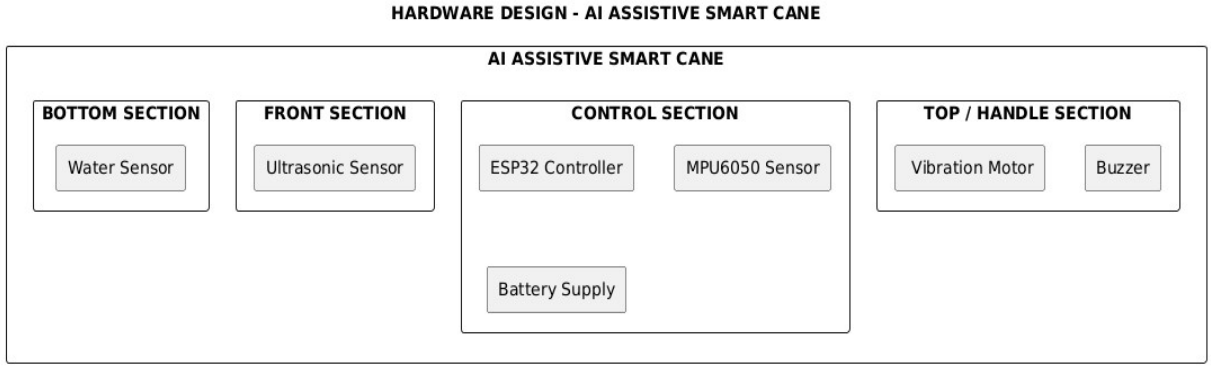
4.1 Circuit Diagrams / PCB Layouts

The AI Assistive Smart Cane circuit is developed by connecting different electronic components to the ESP32 microcontroller. The ultrasonic sensor is connected to detect obstacles, while the water sensor is used to identify wet surfaces. The MPU6050 is connected to monitor the movement and orientation of the cane. The buzzer and vibration motors are connected to provide sound and vibration alerts based on sensor inputs. All components are connected using jumper wires and tested to ensure proper working of the system.



4.2 Hardware Design

The hardware design of the AI Assistive Smart Cane consists of an ESP32 microcontroller connected to an ultrasonic sensor, water sensor, MPU6050, buzzer, and vibration motors. The ultrasonic sensor is placed at the front of the cane to detect obstacles, while the water sensor is placed at the bottom to identify wet surfaces. The MPU6050 is used to monitor the movement and orientation of the cane. The buzzer and vibration motors are arranged to provide sound and vibration alerts to the user. All components are connected using jumper wires and powered through a battery supply. The overall hardware arrangement is designed to be simple, lightweight, affordable, and suitable for daily use.



4.3 Software Design

The software design of the AI Assistive Smart Cane is developed using Arduino IDE with Embedded C/C++ programming. The ESP32 microcontroller is programmed to receive data from the ultrasonic sensor, water sensor, and MPU6050. Based on the sensor readings, the program controls the buzzer and vibration motors. The buzzer provides different sound alerts depending on the distance of nearby obstacles, while the vibration motors alert the user when water is detected. The software continuously monitors the sensors to ensure proper functioning of the system.

Code:

```
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>
#include <Wire.h>

// ---------- BLE ----------
#define DEVICE_NAME  "ESP32 Safety Monitor"
#define SERVICE_UUID "4fafc201-1fb5-459e-8fcc-c5c9c331914b"
#define CHAR_UUID    "beb5483e-36e1-4688-b7f5-ea07361b26a8"

// ---------- Pins ----------
#define TRIG_PIN      5
#define ECHO_PIN      18   // use a voltage divider (HC-SR04 echo is 5V)
#define WATER_PIN     27
#define BUZZER_PIN    26
#define VIBRATOR_PIN  25   // vibrator 1, drive through a transistor
#define VIBRATOR2_PIN 33   // vibrator 2, drive through its own transistor
#define MPU_SDA       21
#define MPU_SCL       22

// ---------- Settings ----------
#define DIST_THRESHOLD_CM   25
#define VIBRATOR_TIME_MS    3000
#define WATER_ACTIVE_STATE  HIGH   // change to LOW for active-low modules
#define GYRO_LIMIT_DPS      60.0
#define ACCEL_LIMIT_G       0.35
```

```
#define NOTIFY_EVERY_MS     200

const uint8_t MPU_ADDR = 0x68;

BLEServer* bleServer = nullptr;
BLECharacteristic* dataChar = nullptr;
bool clientConnected = false;
bool wasConnected = false;

// ---------- Latest readings ----------
long  distanceCm = -1;
bool  objectNear = false, waterDetected = false, unstable = false;
bool  buzzerOn = false, vibratorOn = false, lastObjectNear = false;
float gyroMag = 0, accMag = 1;
unsigned long vibratorStart = 0, lastNotify = 0;

class ServerCallbacks : public BLEServerCallbacks {
  void onConnect(BLEServer*)    { clientConnected = true;  Serial.println("Client connected"); }
  void onDisconnect(BLEServer*) { clientConnected = false; Serial.println("Client disconnected");
}
};

long readDistanceCm() {
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);
  long duration = pulseIn(ECHO_PIN, HIGH, 30000);
  if (duration == 0) return -1;
  return duration * 0.0343 / 2;
}

void mpuInit() {
  Wire.begin(MPU_SDA, MPU_SCL);
```

```
  Wire.beginTransmission(MPU_ADDR);
  Wire.write(0x6B);
  Wire.write(0);
  Wire.endTransmission(true);
}

bool readMpu() {
  Wire.beginTransmission(MPU_ADDR);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(MPU_ADDR, (uint8_t)14, (uint8_t)true);
  if (Wire.available() < 14) return true;   // no response = unstable

  int16_t ax = Wire.read() << 8 | Wire.read();
  int16_t ay = Wire.read() << 8 | Wire.read();
  int16_t az = Wire.read() << 8 | Wire.read();
  Wire.read(); Wire.read();
  int16_t gx = Wire.read() << 8 | Wire.read();
  int16_t gy = Wire.read() << 8 | Wire.read();
  int16_t gz = Wire.read() << 8 | Wire.read();

  float axg = ax / 16384.0, ayg = ay / 16384.0, azg = az / 16384.0;
  float gxd = gx / 131.0,   gyd = gy / 131.0,   gzd = gz / 131.0;
  accMag  = sqrt(axg * axg + ayg * ayg + azg * azg);
  gyroMag = sqrt(gxd * gxd + gyd * gyd + gzd * gzd);
  return (fabs(accMag - 1.0) > ACCEL_LIMIT_G) || (gyroMag > GYRO_LIMIT_DPS);
}

// 9-byte packet: flags, distance(int16), vibratorLeft(uint16), gyro x10(uint16), accel x100(uint16)
void sendPacket() {
  uint16_t left = 0;
  if (vibratorOn) {
    unsigned long used = millis() - vibratorStart;
    left = used < VIBRATOR_TIME_MS ? VIBRATOR_TIME_MS - used : 0;
  }
```

```
  uint8_t flags = (objectNear ? 1 : 0) | (waterDetected ? 2 : 0) | (unstable ? 4 : 0) |
                  (buzzerOn ? 8 : 0) | (vibratorOn ? 16 : 0);
  int16_t  d = (int16_t)distanceCm;
  uint16_t g = (uint16_t)constrain(gyroMag * 10, 0, 65535);
  uint16_t a = (uint16_t)constrain(accMag * 100, 0, 65535);

  uint8_t p[9];
  p[0] = flags;
  p[1] = d & 0xFF;     p[2] = (d >> 8) & 0xFF;
  p[3] = left & 0xFF;  p[4] = (left >> 8) & 0xFF;
  p[5] = g & 0xFF;     p[6] = (g >> 8) & 0xFF;
  p[7] = a & 0xFF;     p[8] = (a >> 8) & 0xFF;

  dataChar->setValue(p, 9);
  dataChar->notify();
}

void setVibrators(bool on) {
  digitalWrite(VIBRATOR_PIN,  on ? HIGH : LOW);
  digitalWrite(VIBRATOR2_PIN, on ? HIGH : LOW);
}

void setup() {
  Serial.begin(115200);
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(WATER_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(VIBRATOR_PIN, OUTPUT);
  pinMode(VIBRATOR2_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);
  setVibrators(false);
  mpuInit();

  BLEDevice::init(DEVICE_NAME);
```

```
  bleServer = BLEDevice::createServer();
  bleServer->setCallbacks(new ServerCallbacks());

  BLEService* service = bleServer->createService(SERVICE_UUID);
  dataChar = service->createCharacteristic(CHAR_UUID,
          BLECharacteristic::PROPERTY_READ | BLECharacteristic::PROPERTY_NOTIFY);
  dataChar->addDescriptor(new BLE2902());
  service->start();

  BLEAdvertising* adv = BLEDevice::getAdvertising();
  adv->addServiceUUID(SERVICE_UUID);
  adv->setScanResponse(true);
  adv->setMinPreferred(0x06);
  adv->setMinPreferred(0x12);
  BLEDevice::startAdvertising();
  Serial.println("Advertising as: " DEVICE_NAME);
  Serial.print("Bluetooth address: ");
  Serial.println(BLEDevice::getAddress().toString().c_str());
}

void loop() {
  distanceCm    = readDistanceCm();
  objectNear    = (distanceCm > 0 && distanceCm <= DIST_THRESHOLD_CM);
  waterDetected = (digitalRead(WATER_PIN) == WATER_ACTIVE_STATE);
  unstable      = readMpu();   // still sent to the dashboard, does not drive the buzzer

  // Buzzer: ON if object within 25 cm OR water detected, otherwise OFF
  buzzerOn = objectNear || waterDetected;
  digitalWrite(BUZZER_PIN, buzzerOn ? HIGH : LOW);

  // Both vibrators: ON for 3 s when the ultrasonic detects an object
  if (objectNear && !lastObjectNear) {
    vibratorOn = true;
    vibratorStart = millis();
    setVibrators(true);
```

```
  }
  lastObjectNear = objectNear;

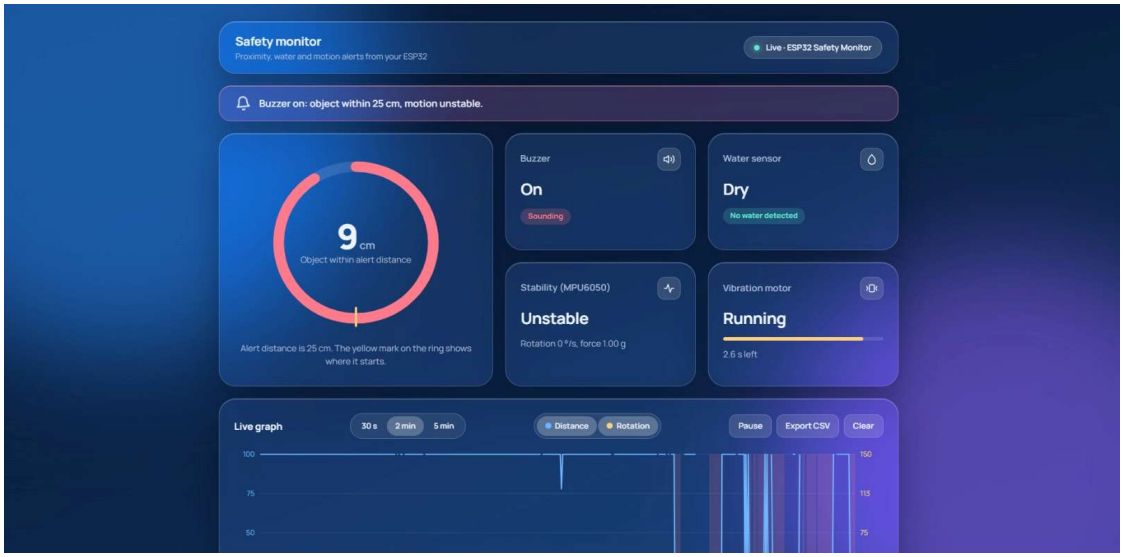
  if (vibratorOn && millis() - vibratorStart >= VIBRATOR_TIME_MS) {
    vibratorOn = false;
    setVibrators(false);
  }

  // Send readings to the dashboard
  if (clientConnected && millis() - lastNotify >= NOTIFY_EVERY_MS) {
    lastNotify = millis();
    sendPacket();
  }

  // Start advertising again after the phone/computer disconnects
  if (!clientConnected && wasConnected) {
    delay(300);
    BLEDevice::startAdvertising();
  }
  wasConnected = clientConnected;

  delay(50);
}
```

HTML Interface



4.4 Communication Protocols Used

The AI Assistive Smart Cane uses communication protocols to transfer data between the sensors and the ESP32 microcontroller. The MPU6050 communicates with the ESP32 using the I2C protocol through SDA and SCL pins. The ultrasonic sensor uses digital GPIO pins to send trigger signals and receive echo signals for distance measurement. The water sensor sends its output signal to the ESP32 through a GPIO pin. The buzzer and vibration motors are controlled using digital output signals from the ESP32. These communication methods help the components work together and provide proper warning alerts to the user.

4.5 System Integration Details

The AI Assistive Smart Cane integrates different hardware and software components using the ESP32 microcontroller as the main controlling unit. The ultrasonic sensor, water sensor, and MPU6050 are connected to the ESP32 to collect information about obstacles, wet surfaces, and cane movement. The sensor data is processed using the programmed logic, and the buzzer and vibration motors provide alerts based on the detected conditions. The HTML interface is also developed to present the project information through a web-based display. All the components are integrated and tested to ensure proper communication and smooth operation of the system .

CHAPTER – 5

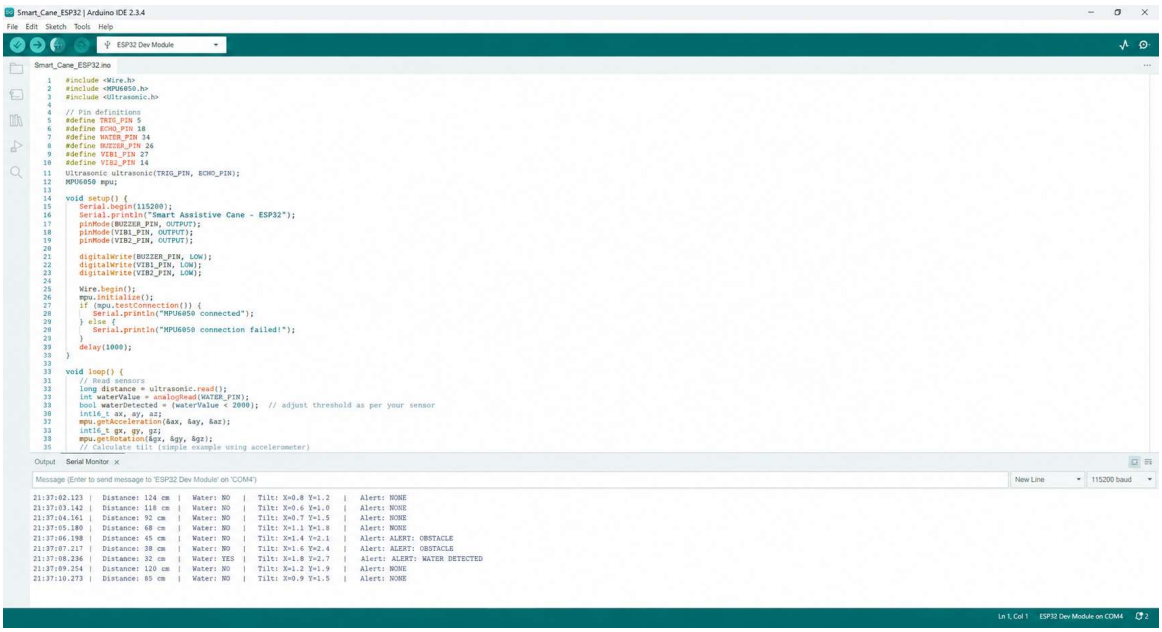
RESULTS AND DISCUSSION

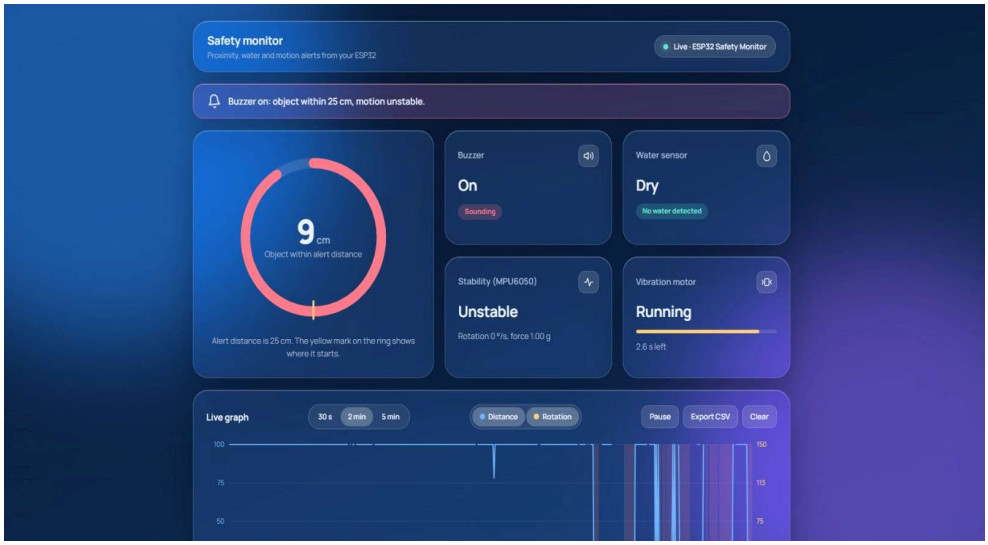
5.1 Experimental Setup

The experimental setup of the AI Assistive Smart Cane consists of an ESP32 microcontroller connected to an ultrasonic sensor, water sensor, MPU6050, buzzer, and vibration motors. All the components are connected using jumper wires and powered through a battery supply. The ultrasonic sensor is tested by placing objects at different distances to check obstacle detection and buzzer alerts. The water sensor is tested using water to verify the vibration response. The MPU6050 is checked by moving and tilting the cane to observe its movement readings. The complete system is tested to verify the proper functioning of all components.

5.2 Output Results

The AI Assistive Smart Cane provides obstacle detection, water detection, and motion monitoring using the connected sensors and ESP32 microcontroller. When an obstacle is detected within 30 cm, the buzzer produces two beeps, and when the obstacle comes closer than 27 cm, the buzzer beeps continuously. When water is detected, the vibration motors provide a tactile alert to the user. The MPU6050 monitors the movement and position of the cane. These outputs demonstrate how sensor inputs can be used to provide sound and vibration warnings to assist visually impaired people while walking. The system aims to improve obstacle awareness and walking safety through simple and affordable electronic components.





5.3 Performance Analysis

The performance of the AI Assistive Smart Cane is evaluated by testing its sensors and alert systems under different conditions. The ultrasonic sensor detects nearby obstacles and activates the buzzer with two beeps when an object comes within 30 cm and continuous beeping when it comes closer than 27 cm. The water sensor successfully detects water or wet surfaces and activates the vibration motors to alert the user. The MPU6050 monitors the movement and orientation of the cane. The ESP32 processes the sensor readings and controls the output devices. Overall, the system is designed to provide quick warnings through sound and vibration, improving the safety and convenience of visually impaired people.

CHAPTER – 6

CONCLUSION AND FUTURE SCOP

6.1 Conclusion

The AI Assistive Smart Cane project is developed to improve the safety and independence of blind and visually impaired people during walking. The system combines an ESP32 microcontroller with an ultrasonic sensor, water sensor, MPU6050, buzzer, and vibration motors to provide assistance through sound and vibration alerts. The ultrasonic sensor detects obstacles in front of the user and activates the buzzer based on the distance. When an object comes within 30 cm, the buzzer produces two beeps, and when it comes closer than 27 cm, it produces continuous beeping. Similarly, the water sensor detects water or wet surfaces and activates the vibration motors to alert the user through touch. The MPU6050 is used to monitor the movement and orientation of the cane.

The integration of these components makes the traditional walking cane more useful by providing additional information about the user's surroundings. The ESP32 controls the entire system and processes the sensor readings to generate suitable alerts. An HTML-based interface is also developed as part of the software component of the project. The system uses simple, affordable, and easily available electronic components, making it suitable for a low-cost implementation.

Overall, the project demonstrates how technology can be used to improve the daily lives of visually impaired people. It provides an additional level of assistance by warning users about nearby obstacles and wet surfaces. Although the system has some limitations, it serves as a useful step towards developing smarter and more effective mobility assistance devices. The main goal of the project is to help visually impaired people move with greater confidence, safety, and independence.

6.2 Future Scope

The AI Assistive Smart Cane can be further improved by adding advanced features to provide better assistance to blind and visually impaired people. In the future, a GPS module can be integrated to help users identify their current location and navigate to different places. Voice guidance can also be added to provide audio instructions about nearby obstacles and directions. A camera module with AI-based image recognition can be included to identify objects, people, vehicles, and other surroundings. Wi-Fi or Bluetooth connectivity can be used to send emergency alerts and location details to family members. A mobile application can also be developed to monitor the cane's status and provide additional support. The battery system can be improved to provide longer operating time and convenient charging. These improvements can make the AI Assistive Smart Cane more intelligent, reliable, and useful for everyday activities. Overall, future developments aim to provide greater safety, independence, and confidence to visually impaired people

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- Research articles and technical resources on smart canes, assistive technologies, and obstacle detection for visually impaired people..